“Layer su layer”: Identifying and Disambiguating the Italian NPN Construction in BERT's family

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MAIN CONTRIBUTION: We investigate whether contextual embeddings in BERT encode constructional (form-meaning) information, using the Italian NPN Cxn as a test case. Through probing experiments, we show that models capture both structural and semantic properties, supporting a constructionist view of language representation.

CxG and PLMs

Interpretability research aims to understand whether the **internal representations** of PLMs encode linguistically meaningful structure.

Cxns are ideal test cases for interpretability [1], as they encode **form–meaning pairings** at the **syntax–semantics interface** [2].

As exposed in a previous study on English [3], **NPN Cxn** is particularly informative: it combines formal constraints with multiple competing meanings and forms a network of **semi-schematic patterns** [4].

This makes it a challenging setting to test whether **PLMs capture structured constructional information**, beyond surface-level lexical cues.

Formal constraints (reduplication, absence of determiners)
Cxn-specific meanings (e.g., succession, contact, accumulation)

The **dataset** [5] is derived from the Italian NPN resource [6] and extended with full sentential contexts.

Data and methods

For this study, the analysis is restricted to NPN Cxn (1-4) with the prepositions *a* and *su*

Distractors counterpart of the dataset [5]:
- **surface-isomorphic pattern**
- **structurally distinct Cxns**.

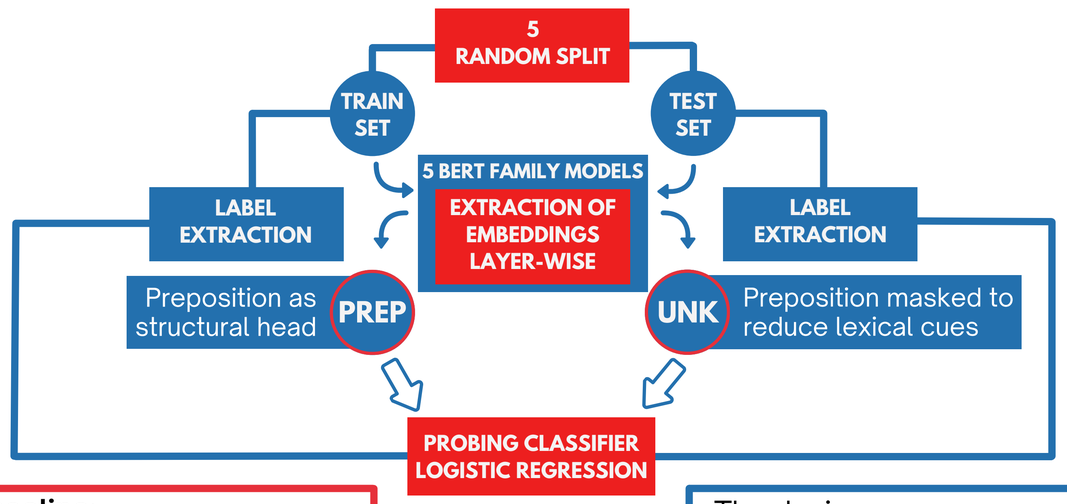
This allows the task to move beyond purely syntactic discrimination and test whether models capture true constructional status, i.e., form–meaning pairings rather than superficial patterns.

	Form	Meaning	Types	Tokens
1	$[[x]_{N_i}$ *a* $[x]_{N_i}]_{\text{MOD}_j}$	Succession, iteration, distributivity	21	399
2	$[[x]_{N_i}$ *a* $[x]_{N_i}]_{\text{MOD}_j}$	Juxtaposition, contact	27	1108
3	$[[x]_{N_i^{sg}}$ *su* $[x]_{N_i^{sg}}]_{\text{MOD}_j}$	Succession, iteration, distributivity	71	178
4	$[[x]_{N_i^{pl}}$ *su* $[x]_{N_i^{pl}}]_{\text{MOD}_j}$	Greater plurality, accumulation	252	403
5	$[[x]_{N_i}$ *per* $[x]_{N_i}]_{\text{MOD}_j}$	Succession, iteration, distributivity	385	3324
6	$[[x]_{N_i}$ *per* $[x]_{N_i}]_{\text{SUB.CL}_j}$	Inescapable presupposition	21	21
7	$[[x]_{N_i}$ *dopo* $[x]_{N_i}]_{\text{MOD}_j}$	Succession, iteration, distributivity	368	2387
8	$[[x]_{N_i}$ *contro* $[x]_{N_i}]_{\text{MOD}_j}$	Juxtaposition, contact	56	170

DISTRACTORS 989 + CONSTRUCTIONS 1281 → FULL DATASET

Example:
(...)tre donne che scrivono per ***donne su donne****(...)*
‘(...)three women who write for **women on women**(...)’

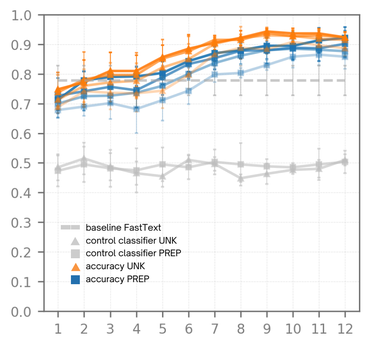
Example:
(...)si battono ***testa a testa*** *in un torneo(...)*
‘(...)they battle **head to head** in the round(...)’



Baseline:
- Classifier with **static embeddings** (FastText [8])
- **Control Classifier** [7]

The design ensures performance reflects encoded information, not classifier complexity.

Can BERT distinguish NPN Cxn from distractors?



High accuracy in middle-late layers

UNK ≈ PREP comparable performance with and without lexical cues

Models capture **construction-level structure**, not just lexical cues

FORM OF THE CXN

Identification as a binary classification task
- Positive: NPN Cxns
- Negative: distractors

This tests whether models capture **constructional status**

Can BERT distinguish NPN’s construction-specific meanings?

MEANING OF THE CXN

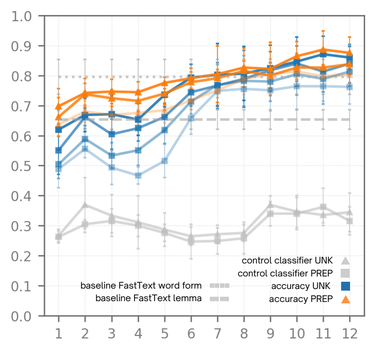
Disambiguation as a multi-class classification task
- Succession
- Accumulation
- Contact

This tests whether models capture **constructional meaning**

Performance improves across layers

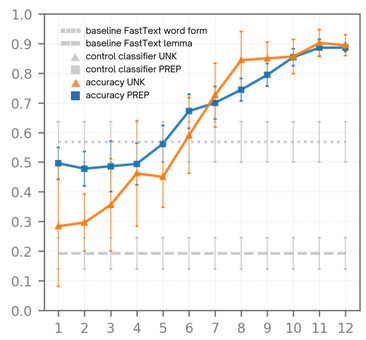
Higher layers encode Cxn’s meaning

Confusions reflect meaning overlap



Models encode **constructional meaning**, but distinctions remain gradient

Semantic generalisation



Strong improvement in higher layers

UNK > PREP in early layers

Models encode **abstract constructional meaning**, not tied to specific prepositions

GENERALISATION TO UNSEEN PREPOSITIONS

TRAIN SET: Distractors + Cxns (1-4)

TEST SET: Cxns (5,7), instanciated by *per* and *dopo*

This tests whether models capture **abstraction** beyond **lexical realisation**

Future directions

⚠ Results remain confined to the representational space within BERT’s models and theoretical implications should be interpreted with caution.

Probing results should be cross-validated with behavioural testing experiments

Empirical ground for theoretical debate on NPN Cxn’s schematisation

Direct comparison between human judgments with model’s predictions

References:

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[3] Scivetti, W., & Schneider, N. (2025). Construction identification and disambiguation using BERT: A case study of NPN. In Proc. CoNLL 2025, 365–376.
[4] Masini, F. (2024). Costruzioni su costruzioni: Idiomaticity and regularity of NPN discontinuous reduplications in Italian. In I fraseologismi schematici, 51–82.
[5] Gorzoni, G., Pannitto, L., & Masini, F. (2026). NPN Construction and Distractor Dataset.
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[7] Hewitt, J., & Liang, P. (2019). Designing and interpreting probes with control tasks. In Proc. EMNLP-IJCNLP 2019, 2733–2743.
[8] Bojanowski, P., Grave, E., Joulin, A., & Mikolov, T. (2017). Enriching word vectors with subword information. In Proc. ACL 2017, 135–146.